

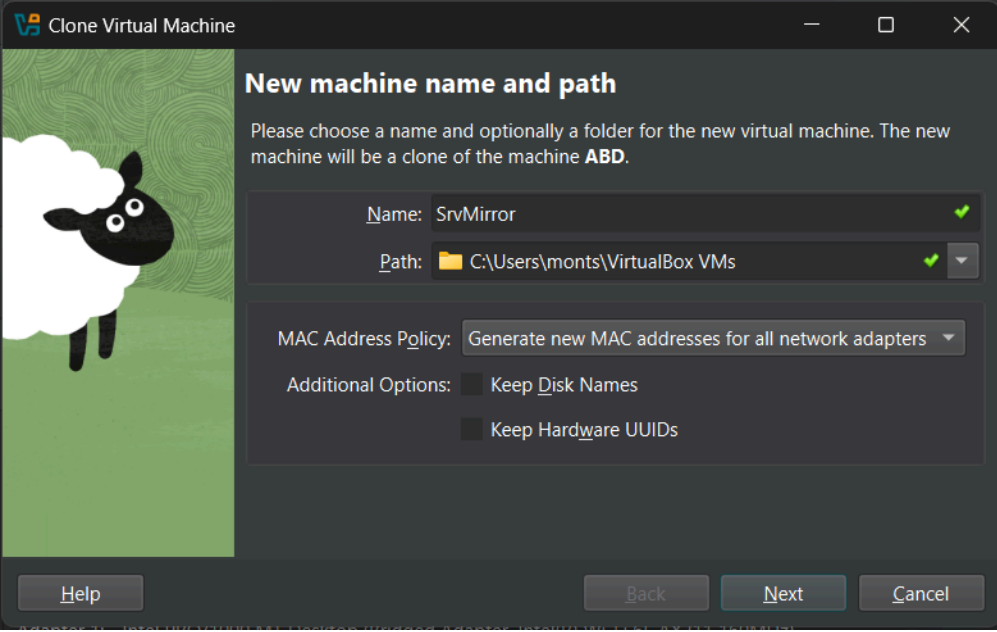
```
mont89@ABD:~$ sudo systemctl status postgresql
[sudo] password for mont89:
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Mon 2026-05-11 08:04:33 MDT; 2min 13s ago
   Process: 853 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 853 (code=exited, status=0/SUCCESS)
   CPU: 11ms

may 11 08:04:33 ABD systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
may 11 08:04:33 ABD systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
```

```
mont89@ABD:~$ sudo truncate -s 0 /etc/machine-id
mont89@ABD:~$ sudo rm -f /var/lib/dbus/machine-id
mont89@ABD:~$ sudo ln -s /etc/machine-id /var/lib/dbus/machine-id
mont89@ABD:~$ cd /etc
mont89@ABD:/etc$ ls
adduser.conf      dhcp              landscape         multipath.conf    rc1.d             sudo_logsrvd.conf
alternatives      dhcpcd.conf      ldap              nanorc            rc2.d             supercat
apparmor          dpkg             ld.so.cache      needrestart       rc3.d             sysctl.conf
apparmor.d        e2scrub.conf    ld.so.conf       netconfig        rc4.d             sysctl.d
appport           environment      ld.so.conf.d     netplan          rc5.d             sysstat
apt               ethertypes      legal            network          rc6.d             systemd
bash.bashrc       fonts           libaudit.conf   networkd-dispatcher rcS.d             terminfo
bash_completion   fstab           libblockdev     networks         resolv.conf       thermald
bash_completion.d fuse.conf        libibverbs.d    newt             rmt               timezone
bindresvport.blacklist fwupd           locale.alias    nftables.conf   rpc               tmpfiles.d
binfmt.d          gai.conf        locale.conf     nsswitch.conf   rsyslog.conf     ubuntu-advantage
byobu             gnupg           locale.gen      os-release      rsyslog.d        ucf.conf
ca-certificates   groff           localtime       overlayroot.conf security          udev
ca-certificates.conf group           logcheck        PackageKit      selinux          udisks2
cloud             group.d         logrotate.conf  pam.conf        sensors3.conf    ufw
console-setup     gshadow         logrotate.d     pam.d           sensors.d        update-manager
credstore         gshadow-       lsb-release     passwd          services         update-motd.d
credstore.encrypted gss            lvm             passwd-         sgml             update-notifier
cron.d            hdparm.conf    machine-id      perl            shadow           UPower
cron.daily        host.conf      magic           pki             shadow-         usb_modeswitch.conf
cron.hourly       hostname       magic           plymouth        shells           usb_modeswitch.d
cron.monthly      hostnames      magic           plymouth        shells           vconsole.conf
```

```
mont89@ABD:/etc$ ls machine-id
machine-id
mont89@ABD:/etc$ cat machine-id
mont89@ABD:/etc$ |
```

```
mont89@ABD:~$ sudo rm -f /etc/ssh/ssh_host_*
mont89@ABD:~$ sudo shutdown now
```



Clone Virtual Machine

New machine name and path

Please choose a name and optionally a folder for the new virtual machine. The new machine will be a clone of the machine **ABD**.

Name: ✓

Path: ✓

MAC Address Policy:

Additional Options: ☐ Keep Disk Names ☐ Keep Hardware UUIDs

Help Back Next Cancel

```

mont89@ABD:~$ sudo systemd-machine-id-setup
[sudo] password for mont89:
mont89@ABD:~$ cd /etc
mont89@ABD:/etc$ cat machine-id
ddeb5c9c98848218812c6c2b4c84496
mont89@ABD:/etc$ sudo dpkg-reconfigure openssh-server
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:AAC0s7C06EbNhnZF7Fj6lSLcnEQKfo0pGVeNK2Fdrn0 root@ABD (RSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:FifJNHref70fEfjFLihxcjZxo+e8agqxHpjeLn0yBq4 root@ABD (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:+CzcXzFQ6uHrBRWXGhBKDCTLC86SuarzmRJlpeOYRZY root@ABD (ED25519)

```

```

C:\Users\monts>ssh mont89@192.168.1.224
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!    @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:+CzcXzFQ6uHrBRWXGhBKDCTLC86SuarzmRJlpeOYRZY.
Please contact your system administrator.
Add correct host key in C:\Users\monts\.ssh\known_hosts to get rid of this message.
Offending ED25519 key in C:\Users\monts\.ssh\known_hosts:14
Host key for 192.168.1.224 has changed and you have requested strict checking.
Host key verification failed.

```

```

C:\Users\monts>ssh-keygen -R 192.168.1.224
# Host 192.168.1.224 found: line 14
C:\Users\monts\.ssh\known_hosts updated.
Original contents retained as C:\Users\monts\.ssh\known_hosts.old

```

```

C:\Users\monts>ssh mont89@192.168.1.224
The authenticity of host '192.168.1.224 (192.168.1.224)' can't be established.
ED25519 key fingerprint is SHA256:+CzcXzFQ6uHrBRWXGhBKDCTLC86SuarzmRJlpeOYRZY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.224' (ED25519) to the list of known hosts.
mont89@192.168.1.224's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

```

```

mont89@ABD:~$ sudo hostnamectl set-hostname SrvPrincipal
[sudo] password for mont89:
mont89@ABD:~$ sudo reboot

```

```

Broadcast message from root@ABD on pts/1 (Mon 2026-05-11 08:20:12 MDT):

The system will reboot now!

```

```

mont89@SrvPrincipal:~$ hostname
SrvPrincipal

```

```

mont89@ABD:~$ sudo systemd-machine-id-setup
[sudo] password for mont89:
mont89@ABD:~$ cd /etc
mont89@ABD:/etc$ cat machine-id
620fa53f421a4af2ad6db530a79acfb1
mont89@ABD:/etc$ cd
mont89@ABD:~$ sudo dpkg-reconfigure openssh-server
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:RG5CovK1B9ZEEvR4DPXJskos7FtALee3R4vpyCfeuoI root@ABD (RSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:yEuT20p/FHjV/FXfdtkqzV4GjDgxnc3LWlfnywEDRy4 root@ABD (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:u3hh/19XcoAD2zXjf8FHcmQoHOJABcnQU+9xbDms7n8 root@ABD (ED25519)

```

```

mont89@ABD:~$ sudo hostnamectl set-hostname SrvEspejo
[sudo] password for mont89:
mont89@ABD:~$ sudo reboot

```

Broadcast message from root@ABD on pts/1 (Mon 2026-05-11 08:25:46 MDT):

The system will reboot now!

```

mont89@SrvEspejo:~$ hostname
SrvEspejo

```

```

mont89@SrvPrincipal:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state U
P group default qlen 1000
    link/ether 08:00:27:a8:81:d7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.224/24 metric 100 brd 192.168.1.255 scope global dynamic
enp0s3
        valid_lft 42735sec preferred_lft 42735sec
    inet6 2806:1016:d:6cf0:a00:27ff:fee8:81d7/64 scope global dynamic mngtmp
addr noprefixroute
        valid_lft 4294492784sec preferred_lft 4294492784sec
    inet6 fe80::a00:27ff:fee8:81d7/64 scope link
        valid_lft forever preferred_lft forever

```

```

mont89@SrvEspejo:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state U
P group default qlen 1000
    link/ether 08:00:27:f2:f6:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.212/24 metric 100 brd 192.168.1.255 scope global dynamic
enp0s3
        valid_lft 43038sec preferred_lft 43038sec
    inet6 2806:1016:d:6cf0:a00:27ff:fef2:f604/64 scope global dynamic mngtmp
addr noprefixroute
        valid_lft 4294492781sec preferred_lft 4294492781sec
    inet6 fe80::a00:27ff:fef2:f604/64 scope link
        valid_lft forever preferred_lft forever

```

GNU nano 7.2 /etc/netplan/01-netcfg.yaml *

```

network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.224/24
      gateway4: 192.168.1.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 1.1.1.1

```

```

GNU nano 7.2 /etc/netplan/01-netcfg.yaml *
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.212/24
      gateway4: 192.168.1.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 1.1.1.1

```

```

mont89@SrvEspejo:~$ sudo netplan apply

** (generate:1012): WARNING **: 08:33:10.133: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.

** (generate:1012): WARNING **: 08:33:10.136: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1010): WARNING **: 08:33:10.949: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.

** (process:1010): WARNING **: 08:33:10.949: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1010): WARNING **: 08:33:11.178: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.

** (process:1010): WARNING **: 08:33:11.180: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

```

<pre> mont89@SrvPrincipal:~\$ ping 192.168.1.212 PING 192.168.1.212 (192.168.1.212) 56(80) bytes of data: 64 bytes from 192.168.1.212: icmp_seq=1 ttl=64 time=4.46 ms 64 bytes from 192.168.1.212: icmp_seq=2 ttl=64 time=2.31 ms 64 bytes from 192.168.1.212: icmp_seq=3 ttl=64 time=1.99 ms 64 bytes from 192.168.1.212: icmp_seq=4 ttl=64 time=1.15 ms 64 bytes from 192.168.1.212: icmp_seq=5 ttl=64 time=2.13 ms 64 bytes from 192.168.1.212: icmp_seq=6 ttl=64 time=2.61 ms 64 bytes from 192.168.1.212: icmp_seq=7 ttl=64 time=2.92 ms 64 bytes from 192.168.1.212: icmp_seq=8 ttl=64 time=2.93 ms 64 bytes from 192.168.1.212: icmp_seq=9 ttl=64 time=1.35 ms 64 bytes from 192.168.1.212: icmp_seq=10 ttl=64 time=0.964 ms 64 bytes from 192.168.1.212: icmp_seq=11 ttl=64 time=1.92 ms 64 bytes from 192.168.1.212: icmp_seq=12 ttl=64 time=1.68 ms 64 bytes from 192.168.1.212: icmp_seq=13 ttl=64 time=1.92 ms 64 bytes from 192.168.1.212: icmp_seq=14 ttl=64 time=2.34 ms 64 bytes from 192.168.1.212: icmp_seq=15 ttl=64 time=1.94 ms 64 bytes from 192.168.1.212: icmp_seq=16 ttl=64 time=1.95 ms ^C --- 192.168.1.212 ping statistics --- 16 packets transmitted, 16 received, 0% packet loss, time 15053ms rtt min/avg/max/mdev = 0.964/2.159/4.459/0.802 ms mont89@SrvPrincipal:~\$ ^C </pre>	<pre> See the 'Default routes' section of the documentation for more details. mont89@SrvEspejo:~\$ ping 192.168.1.224 PING 192.168.1.224 (192.168.1.224) 56(80) bytes of data: 64 bytes from 192.168.1.224: icmp_seq=1 ttl=64 time=1.81 ms 64 bytes from 192.168.1.224: icmp_seq=2 ttl=64 time=2.50 ms 64 bytes from 192.168.1.224: icmp_seq=3 ttl=64 time=2.86 ms 64 bytes from 192.168.1.224: icmp_seq=4 ttl=64 time=5.86 ms 64 bytes from 192.168.1.224: icmp_seq=5 ttl=64 time=2.07 ms 64 bytes from 192.168.1.224: icmp_seq=6 ttl=64 time=1.49 ms 64 bytes from 192.168.1.224: icmp_seq=7 ttl=64 time=1.92 ms 64 bytes from 192.168.1.224: icmp_seq=8 ttl=64 time=1.78 ms 64 bytes from 192.168.1.224: icmp_seq=9 ttl=64 time=3.14 ms 64 bytes from 192.168.1.224: icmp_seq=10 ttl=64 time=1.31 ms 64 bytes from 192.168.1.224: icmp_seq=11 ttl=64 time=1.55 ms 64 bytes from 192.168.1.224: icmp_seq=12 ttl=64 time=1.47 ms 64 bytes from 192.168.1.224: icmp_seq=13 ttl=64 time=2.46 ms 64 bytes from 192.168.1.224: icmp_seq=14 ttl=64 time=1.38 ms 64 bytes from 192.168.1.224: icmp_seq=15 ttl=64 time=1.75 ms 64 bytes from 192.168.1.224: icmp_seq=16 ttl=64 time=3.18 ms ^C --- 192.168.1.224 ping statistics --- 16 packets transmitted, 16 received, 0% packet loss, time 15518ms rtt min/avg/max/mdev = 1.311/2.281/5.856/1.096 ms </pre>
--	---

```
# - Connection Settings -
listen_addresses = '*'
```

```
mont89@SrvEspejo:~$ sudo -u postgres psql -c "SHOW listen_addresses;"
listen_addresses
-----
*
(1 row)
```

```
# Allow replication connections from localhost, by a user with the
# replication privilege.
local    replication    all                                peer
host     replication    all                                127.0.0.1/32      scram-sha-256
host     replication    all                                ::1/128           scram-sha-256
host     replication    replicador                      192.168.1.212/32  scram-sha-256
host     all             all                                192.168.1.212/32  scram-sha-256
```

```
#-----
# REPLICATION
#-----

# - Sending Servers -

# Set these on the primary and on any standby that will send replication data.

#max_wal_senders = 10          # max number of walsender processes
                                # (change requires restart)
#max_replication_slots = 10    # max number of replication slots
                                # (change requires restart)
#wal_keep_size = 0             # in megabytes; 0 disables
#max_slot_wal_keep_size = -1   # in megabytes; -1 disables
#wal_sender_timeout = 60s      # in milliseconds; 0 disables
#track_commit_timestamp = off  # collect timestamp of transaction commit
                                # (change requires restart)
```

```
#-----
# WRITE-AHEAD LOG
#-----

# - Settings -

#wal_level = replica           # minimal, replica, or logical
                                # (change requires restart)
```

```
# - Standby Servers -

# These settings are ignored on a primary server.

#primary_conninfo = ''         # connection string to sending server
#primary_slot_name = ''        # replication slot on sending server
#hot_standby = on              # "off" disallows queries during recovery
                                # (change requires restart)
```

```
mont89@SrvPrincipal:~$ sudo -u postgres psql
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# CREATE ROLE replicador WITH REPLICATION LOGIN PASSWORD 'Taylor1989';
CREATE ROLE
```

```
postgres=# \du
```

Role name	List of roles Attributes
intruso	
postgres	Superuser, Create role, Create DB, Replication, Bypass RLS
replicador	Replication
vendedor1	
vendedor2	

```
postgres=# SELECT * FROM pg_create_physical_replication_slot('slot_espejo');
```

slot_name	lsn
slot_espejo	

(1 row)

```
postgres=# SELECT slot_name, slot_type, active, restart_lsn  
FROM pg_replication_slots;
```

slot_name	slot_type	active	restart_lsn
slot_espejo	physical	f	

(1 row)

Espejo

```
mont89@SrvEspejo:~$ sudo systemctl stop postgresql
```

```
[sudo] password for mont89:
```

```
mont89@SrvEspejo:~$ sudo systemctl status postgresql
```

```
○ postgresql.service - PostgreSQL RDBMS
```

```
Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; p>
```

```
Active: inactive (dead) since Mon 2026-05-11 09:18:47 MDT; 10s ago
```

```
Duration: 52min 13.330s
```

```
Main PID: 818 (code=exited, status=0/SUCCESS)
```

```
CPU: 7ms
```

```
may 11 08:26:33 SrvEspejo systemd[1]: Starting postgresql.service - Postgre>
```

```
may 11 08:26:33 SrvEspejo systemd[1]: Finished postgresql.service - Postgre>
```

```
may 11 09:18:47 SrvEspejo systemd[1]: postgresql.service: Deactivated succe>
```

```
may 11 09:18:47 SrvEspejo systemd[1]: Stopped postgresql.service - PostgreS>
```

```
lines 1-11/11 (END)
```

```
mont89@SrvEspejo:~$ /var/lib/postgresql/16/main
```

```
-bash: /var/lib/postgresql/16/main: Is a directory
```

```
mont89@SrvEspejo:~$ sudo rm -rf /var/lib/postgresql/16/main/*
```

```

mont89@SrvEspejo:~$ sudo rm -rf /var/lib/postgresql/16/main/*
mont89@SrvEspejo:~$ sudo -u postgres pg_basebackup -h 192.168.1.224 -D /var/lib/postgresql/16/main -U replicador -P -v -R -X stream -C -S slot_espejo
Password:
pg_basebackup: error: directory "/var/lib/postgresql/16/main" exists but is not empty
mont89@SrvEspejo:~$ sudo ls -la /var/lib/postgresql/16/main
total 96
drwx----- 20 postgres postgres 4096 may 11 09:18 .
drwxr-xr-x 3 postgres postgres 4096 feb 27 10:59 ..
drwx----- 9 postgres postgres 4096 abr 21 13:30 base
-rw----- 1 postgres postgres 36 may 11 08:26 current_logfiles
drwx----- 2 postgres postgres 4096 may 11 08:27 global
drwx----- 2 postgres postgres 4096 may 11 08:26 log
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_commit_ts
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_dynshmem
drwx----- 4 postgres postgres 4096 may 11 09:18 pg_logical
drwx----- 4 postgres postgres 4096 feb 27 10:59 pg_multixact
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_notify
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_replslot
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_serial
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_snapshots
drwx----- 2 postgres postgres 4096 may 11 09:18 pg_stat
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_stat_tmp
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_subtrans
drwx----- 2 postgres postgres 4096 mar 11 23:53 pg_tblspc
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_twophase
-rw----- 1 postgres postgres 3 feb 27 10:59 PG_VERSION
drwx----- 3 postgres postgres 4096 abr 23 22:01 pg_wal
drwx----- 2 postgres postgres 4096 feb 27 10:59 pg_xact
-rw----- 1 postgres postgres 88 feb 27 10:59 postgresql.auto.conf
-rw----- 1 postgres postgres 130 may 11 08:26 postmaster.opts
mont89@SrvEspejo:~$ sudo find /var/lib/postgresql/16/main -mindepth 1 -delete
mont89@SrvEspejo:~$ sudo ls -la /var/lib/postgresql/16/main
total 8
drwx----- 2 postgres postgres 4096 may 11 09:23 .
drwxr-xr-x 3 postgres postgres 4096 feb 27 10:59 ..

```

```

mont89@SrvEspejo:~$ sudo -u postgres pg_basebackup -h 192.168.1.224 -D /var/lib/postgresql/16/main -U replicador -P -v -R -X stream -S slot_espejo
Password:
pg_basebackup: initiating base backup, waiting for checkpoint to complete
pg_basebackup: checkpoint completed
pg_basebackup: write-ahead log start point: 1/3A000028 on timeline 1
pg_basebackup: starting background WAL receiver
802256/802256 kB (100%), 4/4 tablespaces
pg_basebackup: write-ahead log end point: 1/3A000138
pg_basebackup: waiting for background process to finish streaming ...
pg_basebackup: syncing data to disk ...
pg_basebackup: renaming backup manifest.tmp to backup_manifest
pg_basebackup: base backup completed

```

```

mont89@SrvEspejo:~$ sudo ls -la /var/lib/postgresql/16/main
total 432
drwx----- 20 postgres postgres 4096 may 11 09:28 .
drwxr-xr-x 3 postgres postgres 4096 feb 27 10:59 ..
-rw----- 1 postgres postgres 227 may 11 09:26 backup_label
-rw----- 1 postgres postgres 343240 may 11 09:28 backup_manifest
drwx----- 8 postgres postgres 4096 may 11 09:28 base
-rw----- 1 postgres postgres 36 may 11 09:26 current_logfiles
drwx----- 2 postgres postgres 4096 may 11 09:28 global
drwx----- 2 postgres postgres 4096 may 11 09:26 log
drwx----- 2 postgres postgres 4096 may 11 09:26 pg_commit_ts
drwx----- 2 postgres postgres 4096 may 11 09:26 pg_dynshmem
drwx----- 4 postgres postgres 4096 may 11 09:28 pg_logical
drwx----- 4 postgres postgres 4096 may 11 09:28 pg_multixact
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_notify
drwx----- 2 postgres postgres 4096 may 11 09:26 pg_replslot
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_serial
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_snapshots
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_stat
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_stat_tmp
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_subtrans
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_tblspc
drwx----- 2 postgres postgres 4096 may 11 09:26 pg_twophase
-rw----- 1 postgres postgres 3 may 11 09:28 PG_VERSION
drwx----- 3 postgres postgres 4096 may 11 09:26 pg_wal
drwx----- 2 postgres postgres 4096 may 11 09:28 pg_xact
-rw----- 1 postgres postgres 429 may 11 09:28 postgresql.auto.conf
-rw----- 1 postgres postgres 0 may 11 09:28 standby.signal

```

```

mont89@SrvEspejo:~$ sudo ls /var/lib/postgresql/16/main/standby.signal
/var/lib/postgresql/16/main/standby.signal

```

```

mont89@SrvEspejo:~$ sudo cat /var/lib/postgresql/16/main/postgresql.auto.conf
# Do not edit this file manually!
# It will be overwritten by the ALTER SYSTEM command.
primary_conninfo = 'user=replicador password=Taylor1989 channel_binding=prefer host=192.168.1.224 port=5432 sslmode=prefer sslcompression=0 sslcertmode=allow sslsnl=1 ssl_min_protocol_version=TLSv1.2 gssencmode=prefer krbsrvname=postgres gssdelegation=0 target_session_attrs=any load_balance_hosts=disable'
primary_slot_name = 'slot_espejo'

```

```
mont89@SrvEspejo:~$ sudo systemctl start postgresql
mont89@SrvEspejo:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Mon 2026-05-11 09:38:38 MDT; 5s ago
     Process: 1651 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 1651 (code=exited, status=0/SUCCESS)
      CPU: 43ms

may 11 09:38:38 SrvEspejo systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
may 11 09:38:38 SrvEspejo systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
```

```
mont89@SrvEspejo:~$ sudo -u postgres psql
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1))
Type "help" for help.
```

```
postgres=# SELECT pg_is_in_recovery();
pg_is_in_recovery
-----
t
(1 row)
```

```
postgres=# CREATE DATABASE prueba;
ERROR:  cannot execute CREATE DATABASE in a read-only transaction
```

Principal

```
postgres=# SELECT client_addr, state, sync_state, sent_lsn, write_lsn, flush_lsn, replay_lsn
FROM pg_stat_replication;
 client_addr | state | sync_state | sent_lsn | write_lsn | flush_lsn | replay_lsn
-----+-----+-----+-----+-----+-----+-----
 192.168.1.212 | streaming | async      | 1/3B000148 | 1/3B000148 | 1/3B000148 | 1/3B000148
(1 row)
```

```
postgres=# SELECT slot_name, slot_type, active, restart_lsn
FROM pg_replication_slots;
 slot_name | slot_type | active | restart_lsn
-----+-----+-----+-----
 slot_espejo | physical | t      | 1/3B000148
(1 row)
```

```
postgres=# CREATE DATABASE bd_replica;
CREATE DATABASE
postgres=# \c bd_replica
You are now connected to database "bd_replica" as user "postgres".
bd_replica=# CREATE TABLE alumnos (
    id SERIAL PRIMARY KEY,
    nombre VARCHAR(80),
    carrera VARCHAR(100)
);
CREATE TABLE
bd_replica=# INSERT INTO alumnos (nombre, carrera) VALUES ('Montserrat Valdez', 'Ingeniería en Sistemas Computacionales'),
('Paola Ibarra', 'Ingeniería Industrial'),
('Fatima Saenz', 'Administración');
INSERT 0 3
bd_replica=# SELECT * FROM alumnos;
 id | nombre | carrera
-----+-----+-----
  1 | Montserrat Valdez | Ingeniería en Sistemas Computacionales
  2 | Paola Ibarra | Ingeniería Industrial
  3 | Fatima Saenz | Administración
(3 rows)
```

Espejo

```
postgres=# \c bd_replica
You are now connected to database "bd_replica" as user "postgres".
bd_replica=# SELECT * FROM alumnos;
 id |      nombre      |      carrera
-----+-----+-----
  1 | Montserrat Valdez | Ingeniería en Sistemas Computacionales
  2 | Paola Ibarra      | Ingeniería Industrial
  3 | Fatima Saenz      | Administración
(3 rows)
```

```
bd_replica=# INSERT INTO alumnos (nombre, carrera)
VALUES
('Movimiento con espejo apagado 1', 'ISC'),
('Movimiento con espejo apagado 2', 'ISC');
ERROR:  cannot execute INSERT in a read-only transaction
```

Con el espejo apagado

```
mont89@SrvEspejo:~$ sudo systemctl stop postgresql
mont89@SrvEspejo:~$ sudo systemctl status postgresql
○ postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; pr
  eset: enabled)
   Active: inactive (dead) since Mon 2026-05-11 13:25:41 MDT; 13s ago
 Duration: 10min 35.022s
  Process: 857 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
 Main PID: 857 (code=exited, status=0/SUCCESS)
    CPU: 1ms

may 11 13:15:06 SrvEspejo systemd[1]: Starting postgresql.service - PostgreS
QL RDBMS...
may 11 13:15:06 SrvEspejo systemd[1]: Finished postgresql.service - PostgreS
QL RDBMS.
may 11 13:25:41 SrvEspejo systemd[1]: postgresql.service: Deactivated succes
sfully.
may 11 13:25:41 SrvEspejo systemd[1]: Stopped postgresql.service - PostgreSQ
L RDBMS.
```

Principal:

```
bd_replica=# INSERT INTO alumnos (nombre, carrera)
VALUES
('Movimiento con espejo apagado 1', 'ISC'),
('Movimiento con espejo apagado 2', 'ISC');
INSERT 0 2
```

```
bd_replica=# UPDATE alumnos
SET carrera = 'ISC Actualizado'
WHERE nombre = 'Paola Ibarra';
UPDATE 1
bd_replica=# SELECT * FROM ALUMNOS;
 id |      nombre      |      carrera
-----+-----+-----
  1 | Montserrat Valdez | Ingeniería en Sistemas Computacionales
  3 | Fatima Saenz      | Administración
  4 | Movimiento con espejo apagado 1 | ISC
  5 | Movimiento con espejo apagado 2 | ISC
  2 | Paola Ibarra      | ISC Actualizado
(5 rows)
```

```
bd_replica=# SELECT client_addr, state, sync_state FROM pg_stat_replication;
 client_addr | state | sync_state
-----+-----+-----
(0 rows)
```

```
bd_replica=# SELECT slot_name, active, restart_lsn FROM pg_replication_slots
;
 slot_name | active | restart_lsn
-----+-----+-----
 slot_espejo | f      | 1/3B436378
(1 row)
```

```
mont89@SrvEspejo:~$ sudo systemctl start postgresql
mont89@SrvEspejo:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; p
   Active: active (exited) since Mon 2026-05-11 13:43:30 MDT; 4s ago
   Process: 1180 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1180 (code=exited, status=0/SUCCESS)
      CPU: 44ms

may 11 13:43:30 SrvEspejo systemd[1]: Starting postgresql.service - Postgre
may 11 13:43:30 SrvEspejo systemd[1]: Finished postgresql.service - Postgre
Lines 1-8/9 (END)
```

```
bd_replica=# SELECT * FROM alumnos;
 id | nombre | carrera
-----+-----+-----
  1 | Montserrat Valdez | Ingeniería en Sistemas Computacionales
  3 | Fatima Saenz | Administración
  4 | Movimiento con espejo apagado 1 | ISC
  5 | Movimiento con espejo apagado 2 | ISC
  2 | Paola Ibarra | ISC Actualizado
(5 rows)
```

Con el principal apagado

```
mont89@SrvPrincipal:~$ sudo systemctl stop postgresql
mont89@SrvPrincipal:~$ sudo systemctl status postgresql
○ postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-05-11 13:46:07 MDT; 4s ago
   Duration: 1.586s
   Process: 1329 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1329 (code=exited, status=0/SUCCESS)
      CPU: 39ms

may 11 13:46:05 SrvPrincipal systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
may 11 13:46:05 SrvPrincipal systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
may 11 13:46:07 SrvPrincipal systemd[1]: postgresql.service: Deactivated successfully.
may 11 13:46:07 SrvPrincipal systemd[1]: Stopped postgresql.service - PostgreSQL RDBMS.
```

```
bd_replica=# SELECT * FROM ALUMNOS WHERE nombre = 'Paola Ibarra';
 id | nombre | carrera
-----+-----+-----
  2 | Paola Ibarra | ISC Actualizado
(1 row)
```

```
bd_replica=# SELECT pg_is_in_recovery();
 pg_is_in_recovery
-----
 t
(1 row)
```